

AARHUS UNIVERSITY

DEPARTMENT OF MATHEMATICS

INTRODUCTION TO ALGEBRA 1

Handin 5

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Goofy ah Math Number 5

Digit sum!

Let's remind ourselves what the digit sum is, so look at 12345 that has the digit sum $1 + 2 + 3 + 4 + 5 = 15$ and generally the a number can be represented as

$$a_n a_{n-1} \dots a_1 a_0 = \sum_{i=0}^n a_i 10^i \quad (1)$$

where the left-side represent the digits of the number, not some operation between some numbers $a_n a_{n-1} \dots a_1 a_0$

Show some $n \in \mathbb{N}$ is $3|n$ iff. the digit sum of n , n_{tsum} is also $3|n_{tsum}$

This is equivalent to show

$$n_{tsum} \equiv 0 \pmod{3} \Leftrightarrow n \equiv 0 \pmod{3}$$

start assuming that $n_{tsum} \equiv 0 \pmod{3}$ and rewriting n_{tsum} using (1) and let

$$n_{tsum} = a_n + a_{n-1} + \dots + a_1 + a_0 \quad (2)$$

the substitute n_{tsum} with the above

$$a_n + a_{n-1} + \dots + a_1 + a_0 \equiv 0 \pmod{3} \quad (3)$$

t

$$10 \equiv 1 \pmod{3}$$

then remember that multiplication is preserved in congruence, so

$$10 = 10^i \equiv 1 \pmod{3} \text{ for } i \in \mathbb{N}$$

and then use lemma 9.2 [1] again, so

$$\begin{aligned} a_i \cdot 10^i &\equiv 0 \cdot 1 \pmod{3} \\ &= a_i \cdot 10^i \equiv 0 \pmod{3} \end{aligned}$$

then apply this to every element in (??) to get

$$\begin{aligned} a_n \cdot 10^n &\equiv 0 \pmod{3} \\ a_{n-1} \cdot 10^{n-1} &\equiv 0 \pmod{3} \\ &\vdots \\ a_1 \cdot 10^1 &\equiv 0 \pmod{3} \\ a_0 \cdot 10^0 &\equiv 0 \pmod{3} \end{aligned}$$

then lastly use lemma 9.2 [1] again, to construct n

$$\sum_{i=0}^n a_i 10^i \equiv \sum_{i=0}^n a_i \cdot 10^i \pmod{3}$$

which is

$$n \equiv 0 \pmod{3}$$

Show some $n \in \mathbb{N}$ is $9|n$ iff. the digit sum of n , n_{tsum} is also $9|n$

let $n \in \mathbb{N}$ and let m be a number that occurs when you shift around the individual numbers in n . Show that $n - m$ is a multiple of 9

References

- [1] Johan P. Hansen. *Tal og polynomier. Begreber, metoder, resultater, kodning og kryptografi*. Aarhus Universitetsforlag, 2018.